

ÖZGÜN ELÇİ

<https://ozgunelci.github.io>
ozgunelci@gmail.com

RESEARCH INTERESTS

- Methodology: Decomposition-based hybrid methods; logic-based Benders decomposition; Data-driven optimization; Stochastic, robust, and distributionally robust optimization; Risk-averse optimization.
- Application Interests: AI for Optimization; Transportation and logistics; Network design; Scheduling and routing; Humanitarian logistics; Fairness in resource allocation; Portfolio optimization.

EDUCATION

Carnegie Mellon University, Pittsburgh, PA 2017 - 2022

Ph.D. in Operations Research

Minor in Machine Learning and Statistics

Dissertation: Essays on Logic-Based Benders Decomposition, Portfolio Optimization, and Fair Allocation of Resources

Advisors: Gérard Cornuéjols, John Hooker

Carnegie Mellon University, Pittsburgh PA 2019

M.Sc. in Operations Research

Sabancı University, İstanbul, Turkey 2016

M.Sc. in Industrial Engineering

Thesis: Chance-Constrained Stochastic Programming Models for Humanitarian Relief Network Design

Advisors: Kerem Bülbül, Nilay Noyan

Bahçeşehir University, İstanbul, Turkey 2014

B.Sc. in Industrial Engineering

Graduation Project: A MIP Formulation for Self Financing Energy Investment Planning

Advisors: Semra Ağralı, Ethem Çanakoglu, Yıldız Arıkan

PROFESSIONAL EXPERIENCE

Amazon Seattle, WA

Applied Scientist

December 2025 - Present

Amazon Seattle, WA

Research Scientist

September 2023 - December 2025

Starbucks
Data Scientist

Seattle, WA
April 2023 - September 2023

Amazon
Research Scientist

Seattle, WA
October 2022 - January 2023

Amazon
Research Scientist Intern

Seattle, WA
May 2021 - August 2021

PUBLICATIONS

BOOKS

1. **Elçi, Ö.**, Hooker, J., Zhang, P., 2025. *The Structure of Fair Solutions: Achieving Fairness in an Optimization Model*, Springer.

REFEREED JOURNAL PUBLICATIONS

1. **Elçi, Ö.**, Noyan, N., Bülbül, K., 2018. Chance-Constrained Stochastic Programming under Variable Reliability Levels with an Application to Humanitarian Relief Network Design, *Computers and Operations Research*, 96, 91-107.
2. **Elçi, Ö.**, Noyan, N., 2018. A Chance-Constrained Two-Stage Stochastic Programming Model for Humanitarian Relief Network Design, *Transportation Research Part B: Methodological*, 108, 55-83.
3. **Elçi, Ö.**, Hooker, J.N., 2022. Stochastic Planning and Scheduling with Logic-Based Benders Decomposition, *INFORMS Journal on Computing*, 34(5):2428-2442.
4. Cornuéjols, G., **Elçi, Ö.**, Patil, V., 2026. Addressing Estimation Errors on Expected Asset Returns through Robust Portfolio Optimization, *Quantitative Finance*, 1-14.

WORK-IN-PROGRESS

- On Logic-Based Benders Decomposition and Sequence-Dependent Scheduling, joint work with John Hooker.

PRESENTATIONS

- A Benders Decomposition Approach for Large-Scale Hub Network Design
 - INFORMS Annual Meeting, Atlanta, 2025.
- Structural Properties of Equitable Solutions, joint work with John Hooker and Peter Zhang.
 - ISAIM, Fort Lauderdale, 2024.
 - INFORMS Annual Meeting, Phoenix, 2023, presented by J. Hooker.

- Portfolio Optimization in the Presence of Estimation Errors on the Expected Asset Returns
 - Cornell Young Researchers Workshop (Poster Presentation), Ithaca, 2021.
- Structural Characteristics of Equitable and Efficient Distributions, joint work with John Hooker and Peter Zhang.
 - IFORS, 2021, presented by J. Hooker.
 - INFORMS Annual Meeting, Anaheim, 2021.
- Equitable Allocation of Scarce Medical Resources, joint work with John Hooker and Peter Zhang.
 - INFORMS Annual Meeting, 2020, presented by J. Hooker.
- A Logic-Based Benders Decomposition Algorithm for Two-Stage Stochastic Planning and Scheduling Problem, joint work with John Hooker.
 - International Conference on Stochastic Programming, Trondheim, 2019.
 - INFORMS Annual Meeting, Seattle, 2019.
- Chance-Constrained Stochastic Programming under Variable Reliability Levels with an Application to Humanitarian Relief Network Design, joint work with Nilay Noyan and Kerem Bülbül.
 - INFORMS Annual Meeting, Seattle, 2016.
- A Chance-Constrained Two-Stage Stochastic Programming Model for Humanitarian Relief Network Design, joint work with Nilay Noyan.
 - 37th National Congress on Operations Research and Industrial Engineering (YAEM 2017), İstanbul, 2017.
 - Third Place in YinzOR 2018 Poster Competition, Pittsburgh, 2018.

HONORS AND AWARDS

- PNC Presidential Fellowship, Tepper School of Business, Carnegie Mellon University, 2022.
- William Larimer Mellon Fellowship, Tepper School of Business, Carnegie Mellon University, 2017 - 2022.
- Excellence in Teaching Award (an annually granted award for teaching assistants), Sabancı University, for the academic year 2015 - 2016.

- TÜBİTAK (Scientific and Technological Research Council of Turkey) Graduate Fellowship (Stipend for 2 years), 2014.
- Graduate Scholarship (Full tuition fee waiver + Stipend for 2 years), Sabancı University, 2014.
- B.S. with High Honor Degree, Bahçeşehir University, 2014.
- Academic Achievement Scholarship (Stipend for 3 years), Bahçeşehir University, 2011.
- Undergraduate Scholarship (Full tuition fee waiver), Bahçeşehir University, 2010.
- High School Scholarship (Full tuition fee waiver), American Collegiate Institute, 2005.

TEACHING

CARNEGIE MELLON UNIVERSITY

- **Instructor, Optimization for Business** Summer 2022
(undergraduate)
- **TA, Optimization Methods in Finance** Spring 2022
(MBA, Instructor: G. Cornuéjols)
- **TA and recitation leader, Optimization for Business** Spring 2019, 2020, 2022
(undergraduate, Instructor: B. Moseley)
- **TA and recitation leader, Optimization for Business** Fall 2019, 2020, 2021
(undergraduate, Instructor: A. Li)
- **TA, Financial Optimization** Fall 2021
(Master of Science in Computational Finance, Instructor: J. Peña)
- **TA and recitation leader, Probability and Statistics** Fall 2019, 2020
(MBA, Instructor: J. Peña)
- **TA, Risk Analytics** Fall 2019
(MBA, Instructor: N. Secomandi)

SABANCI UNIVERSITY

- **TA and recitation leader, Deterministic Models in Operations Research** Fall 2015, Spring 2017
(undergraduate, Instructor: T. Ünlüyurt)
- **TA and recitation leader, Deterministic Models in Operations Research** Spring 2015, 2016, Fall 2016
(undergraduate, Instructor: K. Bülbül)
- **TA, Supply Chain Practice** Fall 2015
(undergraduate, Instructor: M. Kaya)

SERVICE

- Reviewer
 - INFORMS Journal of Computing.
 - INFORMS Journal on Optimization.
 - Transportation Research Part B.
 - Annals of Operations Research.
 - Transportation Research Part E.
 - Computers & Industrial Engineering.
 - CP 2021 (subreviewer).
 - IISE Transactions.
- CMU INFORMS Student Chapter
 - Treasurer, 2018.
 - Co-chair, YinzOR 2018 Conference.
 - Co-chair, YinzOR 2019 Conference.
- Professional Affiliations
 - Institute for Operations Research and Management Science (INFORMS).
 - Stochastic Programming Society (SPS).